

# Data Session 7 Key, part 2: Grading the Guess

84-355 Democracy's Data | Session 7 — Surnames and BISG | Instructor copy — do not distribute

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All Part 5 code is given to students. Grade the interpretation.

Every number below was run against the committed data.

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## The lab in one sentence

BISG is the method Voting Rights Act litigation runs on, it can almost never be checked, and when you check it on 46,031 people whose race is on record it identifies Black voters correctly **40.3% of the time**.

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## The findings

```
rbind(surname = sapply(R, function(r) round(100*mean(sur[v$race==r]==r), 1)),
      bisg     = sapply(R, function(r) round(100*mean(bis[v$race==r]==r), 1)))
```

```
#>           white black hispanic asian aian
#> surname  94.7  16.6      71.4  64.0  0.2
#> bisg     91.1  40.3      65.9  55.3  0.4
```

```
c(overall_surname = round(100*mean(sur==v$race), 1),
   overall_bisg    = round(100*mean(bis==v$race), 1))
```

```
#> overall_surname  overall_bisg
#>           61.3           67.5
```

|                             | white | black       | hispanic | asian | aian |
|-----------------------------|-------|-------------|----------|-------|------|
| <b>Surname only</b> (61.3%) | 94.7  | <b>16.6</b> | 71.4     | 64.0  | 0.2  |
| <b>BISG</b> (67.5%)         | 91.1  | <b>40.3</b> | 65.9     | 55.3  | 0.4  |

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|        | white | black        | hispanic | asian | aian |
|--------|-------|--------------|----------|-------|------|
| change | -3.6  | <b>+23.7</b> | -5.5     | -8.7  | +0.2 |

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## The three things to teach

**1. The overall number is a headcount, not a performance.** 61.3% surname-only sounds workable. Guessing “white” for every voter scores 51.5%, because 51.5% of the voters in the file are white. Overall accuracy on an unbalanced problem mostly reports the balance.

**2. Geography is a trade, not an improvement.** Adding the block gains Black voters 23.7 points and *costs* Asian voters 8.7 and Hispanic voters 5.5. Black residents here are geographically concentrated, so the block carries real information; Asian and Hispanic residents are dispersed, so the block pulls them toward the local majority and overrides a surname signal that was already good. **The +6.2 overall conceals a redistribution,** and students should say who paid.

**3. Even at its best it is wrong more often than right about Black voters.** 40.3%. This is the instrument behind Section 2 litigation, and it is the group Section 2 exists to protect. That is not an argument that BISG is illegitimate — it is an argument for knowing its error rate before relying on it, which is almost never reported.

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## Option A — Calibration

```
tp <- post[cbind(1:nrow(post), match(v$race, R))]
round(sapply(R, function(r) mean(tp[v$race == r])), 2)
```

```
#>   white   black hispanic   asian   aian
#>   0.81    0.43    0.60    0.54    0.01
```

Mean probability the model puts on the **true** answer: white **0.81**, black **0.43**, hispanic 0.60, asian 0.54, aian **0.01**.

**3/4:** Reports the numbers and notes they track the accuracy ranking.

**4/4:** Distinguishes *calibration* from *decision*. For Black voters the model puts 0.43 on the truth — it is not wildly confident and wrong, it is genuinely uncertain, and then `max.col` forces a hard guess anyway. **The information loss is in the decision rule, not only the model.** A student who says the probabilities should be carried forward rather than collapsed to a label has found the thing real practitioners argue about.

**4/4, best:** notices aian at 0.01 — the model assigns essentially no probability to American Indian for anyone, so those 1,119 voters cannot be found by any threshold. Recalibration would not fix that.

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## Option B — Take the geography away

```
Mc <- matrix(colSums(P)/sum(P), nrow(v), 5, byrow = TRUE)
b2 <- R[max.col(S * Mc, ties.method = "first")]
c(overall = round(100*mean(b2 == v$race), 1),
  black   = round(100*mean(b2[v$race=="black"] == "black"), 1))
```

```
#> overall    black
#>      58.8      7.7
```

County-only prior: **58.8% overall, 7.7% for Black voters** — *worse than surname alone* (61.3% / 16.6%).

**3/4:** Reports that accuracy falls.

**4/4:** Explains why it falls *below* surname-only. A county prior is not neutral: it multiplies every voter by the same county composition, which is 53.9% white, so it tilts everyone toward white and destroys information the surname had. **A bad prior is worse than no prior**, and “we controlled for geography” means nothing without asking at what scale.

**4/4, best:** connects to the modifiable areal unit problem — the same data at a different geographic scale gives a different answer, and nothing about the choice is announced in a methods section.

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## Option C — The ambiguous names

```
amb <- apply(S, 1, max) < 0.6
c(n = sum(amb),
  surname = round(100*mean(sur[amb] == v$race[amb]), 1),
  bisg    = round(100*mean(bis[amb] == v$race[amb]), 1))
```

```
#>      n surname    bisg
#> 7429.0    44.2    60.0
```

7,429 voters whose surname gives no group above 60%: surname-only **44.2%**, BISG **60.0%**.

**3/4:** Notes BISG does much better here.

**4/4:** Draws the right conclusion — this is the population BISG is *for*, and on it geography adds 15.8 points of overall accuracy, its largest overall gain anywhere in the lab — more than twice the 6.2 it gains on the full file. (The single largest gain of any kind is still the 23.7 points for Black voters in Part 4; this is the largest gain in a group’s *overall* score.) The method is not uniformly weak; it is weak where the surname was already informative and strong where it was not. That is the opposite of the impression Part 4 leaves, and holding both is the point.

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## The two fallbacks in Part 3

Worth five minutes, because it is the closest thing to a live bug in the course.

- **392 voters** live in a block the census records as empty. The geographic term vanishes.
- **8 voters** have a surname that is 100% one race and a block with zero residents of that race, so every category multiplies to zero. **BISG is undefined for them.**

Without the fallbacks R throws no error: `max.col` returns column 1 on an all-zero row, so all 400 would be silently labelled **white** — and 158 of them are Black. Ask the class what would have happened if nobody had checked. This is the same lesson as Session 1’s silent join failure, with higher stakes.

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## Grading

- **Never penalise code.** It is your code.
- **The move that earns a 4** is refusing to summarise the method with one number. Any answer whose headline is “BISG is 67.5% accurate” has missed the lab.
- **Reward students who hold both halves:** it fails badly for Black voters *and* it works well on ambiguous names. Both are true and they are not in tension.

## Anticipated questions

- “*Should courts stop using it?*” — Not the lab’s claim, and resist making it. The defensible statement is that an error rate this uneven should be reported alongside any estimate derived from it, and usually is not.
- “*Why is it so bad for Black voters?*” — Because the surnames are shared. This is the Williams result from the morning lab, quantified on 16,852 people.
- “*Is Houston County typical?*” — Unknown, and the lab says so. One county, moderately segregated. The shape generalises; the numbers should not be quoted elsewhere.
- “*Why does race appear on a registration form at all?*” — Voting Rights Act compliance. The fourth family’s question — *what was this collected for?* — has an unusually clean answer here.

## Connections

- **Morning lab (surnames)** — run that first. This is its check.
- **Session 3 (§203)** and **Session 14 (RPV)** — both use race estimates in legal tests. Students now know what those estimates cost.
- **Session 2 (the decennial census)** — “any-part Black” versus “Black alone, not Hispanic” is 4,528 people in this county alone.